Enterprise Incident Investigation Guide

Real production scenarios — investigation → remediation → postmortem

SRE · Platform · Security · Aligned to ShadowXLab Ops Console

Format: PDF · ~20 pages
Audience: Beginner → Engineer → Enterprise
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INC-2001 — Black Friday checkout latency spike

Severity: P1 · Service: checkout-api · Duration: 47 min

What happened. p95 latency on /checkout rose from 180 ms to 4.2 s. Error rate stable. Pods Running, CPU at 70%.

Investigation. 1. kubectl top pod -n checkout → CPU normal; memory rising.
2. kubectl describe hpa checkout-api → HPA stuck at maxReplicas=10 because metric-server reported 30s-old values.
3. kubectl -n kube-system logs deploy/metrics-server → connection refused to kubelet on a tainted node.

Remediation. Removed the orphan taint; metrics-server recovered; HPA scaled to 18 replicas; latency returned in 90s.

Lessons / preventions. Add alert on metrics-server scrape errors. Pre-scale checkout-api for known events. Add load-shedding middleware.

INC-2002 — Pod evictions during deploy

Severity: P2 · Service: payments-worker · Duration: 22 min

What happened. Rolling update evicted 6 pods with Reason=Preempted. New replicas Pending.

Investigation. kubectl get events --sort-by=.lastTimestamp shows *FailedScheduling: 0/12 nodes available*. kubectl describe node shows allocatable memory exhausted by a new ML namespace without quotas.

Remediation. Applied ResourceQuota to ml namespace; payments-worker rescheduled.

Lessons / preventions. Mandatory ResourceQuota on every namespace; PriorityClass on payments tier.

INC-2003 — DNS storm

Severity: P1 · Service: cluster-wide · Duration: 12 min

What happened. Application pods report 'lookup api.internal: i/o timeout' bursts.

Investigation. kubectl -n kube-system logs deploy/coredns → 'too many open files'.
Pods using ndots:5 default → each lookup tried 5 search-domain combinations.

Remediation. Bumped CoreDNS replicas to 5 + ulimit; injected dnsConfig.options: [{name: ndots, value: '2'}] via admission webhook.

Lessons / preventions. Add CoreDNS HPA based on DNS QPS metric.

INC-2004 — Windows node mass NotReady

Severity: P1 · Service: legacy-erp on Windows worker pool · Duration: 35 min

What happened. All 6 Windows nodes flipped NotReady within 90 seconds.

Investigation. RDP into node → Get-Service kubelet = Stopped. Event Viewer shows kubelet crashed after a Windows Update rebooted HNS.

Remediation. Restart-Service hns; Restart-Service kubelet; pin Windows updates via WSUS.

Lessons / preventions. Drain Windows pool before Patch Tuesday; add HNS health probe to monitoring.

INC-2005 — Secret exfiltration attempt

Severity: P1 · Security · Service: cluster-wide

What happened. Falco alert: *shell_in_container* on a payments pod from a non-interactive user.

Investigation. kubectl auth can-i --list --as=system:serviceaccount:payments:default → had cluster-wide secret read by mistake.
Audit log: actor pulled all secrets in 4 namespaces.

Remediation. Revoked ServiceAccount token, rotated 38 secrets, applied least-privilege Role, enabled etcd encryption.

Lessons / preventions. Mandate PodSecurityAdmission=restricted; weekly RBAC drift scan; rotate SA tokens quarterly.

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